TRANSAXLE—4F27E—FINAL DRIVE GEARSET RATIO CHANGE—DELAYED SHIFTS AFTER FINAL DRIVE GEARSET RATIO CHANGE OR TRANSAXLE REPLACEMENT	Article No. 01-25-4

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ISSUE

When REBUILDING a transmission with an older gear ratio, and installing a new final drive input and transfer shaft input gear set, an upgrade to the Powertrain Control Module (PCM) calibration is necessary.

When REPLACING a transmission with a transmission containing an upgraded gearset an upgrade to the PCM calibration is needed.

ACTION

An internal gear change took place to the 4F27E automatic transaxle during the month of December 2000. The number of teeth on the final drive input gear and the transfer shaft input gear were changed, and a new calibration is available to match the new gear ratio. If a change is made to the gear ratio, a calibration change is necessary or the vehicle may exhibit delayed shifts. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

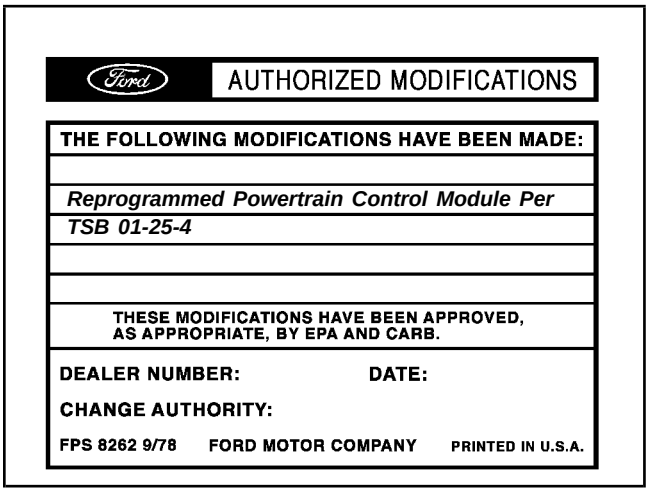
Verify that the vehicle has the old gear ratio. If the vehicle's transaxle has a part number that starts with an X (ex: XS4P) it contains the older gear set. Transaxles with part numbers that begin with 1 (ex: 1S4P) have the new gear set. The old final drive input gear had 87 teeth and the old transfer shaft input gear had 83 teeth. The new final drive input gear has 59 teeth and the new transfer shaft input gear has 57 teeth.

- When REBUILDING a transaxle with a new gear ratio, follow the Workshop Manual procedures to replace the final drive input gear and the transfer shaft input gear assembly. Then reassemble and reprogram the vehicle's PCM to reflect the new gear ratio.
- When REPLACING the transaxle assembly follow the Workshop Manual procedures and reprogram the vehicle's PCM to reflect the new gear ratio.

Refer to the following PCM Information Chart (Figure 1) for details.

Use WDS release B16.3 or later to reprogram PCM.

Obtain an Authorized Modifications Decal (STAT 5941-2 - obtainable through the PDC) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.



OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 501000, 502000, 503000, 503300, 504000

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4F27E Gear Change/Calibration Change

2000 Model Year 4F27E Transaxle With SPI Engine

Use this chart as a guide to upgrade the PCM calibrations when rebuilding or repairing a transmission and using the new gear ratio with the SPI engine.

Rebuild/Replace 4F27E FN Trans Part Number 1S4P-CF (New 3.733 FDR)

Original Calibration	Original PCM	Tear Tag	New Calibration	New PCM	Tear Tag
0AK15S0511	98AB-12A650-AMH	FVV7	0AK15S0511	YU7Z-12A650-DA	OOQ0
0AK15S0512	98AB-12A650-AMJ	FVV8	0AK15S0512	YU7Z-12A650-DA	OOQ0
0AK15S0513	98AB-12A650-AMK	FVV9	0AK15S0513	YU7Z-12A650-DA	OOQ0

2000 Model Year 4F27E Transaxle With Zetec Engine

Use this chart as a guide to upgrade the PCM calibrations when rebuilding or repairing a transmission and using the new gear ratio with the ZETEC engine.

Rebuild/Replace 4F27E FN Trans Part Number 1S4P-DF (New 3.956 FDR)

Original Calibration	Original PCM	Tear Tag	New Calibration	New PCM	Tear Tag
0AK15Z0511	98AB-12A650-AHH	LDC7	0AK15Z0511	YU7Z-12A650-DVA	INU0
0AK15Z0512	98AB-12A650-AHJ	LDC8	0AK15Z0512	YU7Z-12A650-DVA	INU0

2001 Model Year 4F27E Transaxle With SPI Engine

Use this chart as a guide to upgrade the PCM calibrations when rebuilding or repairing a transmission and using the new gear ratio with the SPI engine.

Rebuild/Replace 4F27E FN Trans Part Number 1S4P-CF (New 3.733 FDR)

Original Calibration	Original PCM	Tear Tag	New Calibration	New PCM	Tear Tag
1AK1AS0507	1S4F-12A650-LD	BVP3	1AK1AS0516	1M5Z-12A650-MB	QOW1
1AK1AS0510	1S4F-12A650-LE	BVP4	1AK1AS0516	1M5Z-12A650-MB	QOW1
1AK1AS0515	1M5F-12A650-MA	QOW0	1AK1AS0516	1M5Z-12A650-MB	QOW1
1AK1AS0A06	1S4F-12A650-ACC	RXV2	1AK1AS0A16	1M5Z-12A650-NB	IBE1
1AK1AS0A10	1S4F-12A650-ACD	RXV3	1AK1AS0A16	1M5Z-12A650-NB	IBE1
1AK1AS0A15	1M5F-12A650-NA	IBE0	1AK1AS0A16	1M5Z-12A650-NB	IBE1

2001 Model Year 4F27E Transaxle With Zetec Engine

Use this chart as a guide to upgrade the PCM calibrations when rebuilding or repairing a transmission and using the new gear ratio with the ZETEC engine.

Rebuild/replace 4F27E FN Trans Part Number 1S4P-DF (New 3.956 FDR)

Original Calibration	Original PCM	Tear Tag	New Calibration	New PCM	Tear Tag
1AK1AZ0509	1S4F-12A650-MF	JFR5	1AK1AZ0510	1U7Z-12A650-EA	KBM0
1AK1AZ0510	1S4F-12A650-MG	JFR6	1AK1AZ0510	1U7Z-12A650-EA	KBM0
1AK1AZ0515	1S4F-12A650-PA	CJW0	1AK1AZ0516	1M5Z-12A650-PB	CJW1
1AK1AZ0A09	1S4F-12A650-ADF	SZR5	1AK1AZ0A10	1U7Z-12A650-FAA	VVM0
1AK1AZ0A10	1S4F-12A650-ADG	SZR6	1AK1AZ0A10	1U7Z-12A650-FAA	VVM0
1AK1AZ0A15	1M5F-12A650-RA	XAU0	1AK1AZ0A16	1M5Z-12A650-RB	XAU1

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Figure 1 - Article 01-25-4